Hard Sample Meta-Learning for CIR NLOS Identification in UWB Positioning

Yinong Liu, Haonan Si, *Graduate Student Member, IEEE*, Gordon Owusu Boateng, *Member, IEEE*, Xiansheng Guo, *Senior Member, IEEE*, Yu Cao, *Graduate Student Member, IEEE*, Bocheng Qian, and Nirwan Ansari, *Life Fellow, IEEE*

***Abstract*—Nonline-of-sight (NLOS) identification is the key technique to improve the accuracy of the channel impulse response (CIR)-based ultrawideband (UWB) positioning system. However, most existing NLOS identification approaches are tailored to static environments and often encounter difficulties in dynamic settings with both temporal and spatial variations, particularly when dealing with limited and hard samples. This article introduces a hard sample meta-learning (HSML) approach to address the issues of NLOS identification across different scenarios and domains. HSML includes two phases: 1) a hard sample meta-training phase and 2) a fine-grained meta-testing phase. During the meta-training phase, we train a two-loop learning network using CIR from multiple scenarios (tasks). The inner loop focuses on learning task-specific features, while the outer loop captures cross-task generalization properties using a cross-entropy loss. Hard samples are identified based on estimated residuals for each task, and a new dataset is created, consisting of both hard samples and samples with small residuals. To improve the robustness against hard samples, we implement a residual-corrected focal loss, which is used to retrain the network on this new dataset. In the fine-grained meta-testing phase, we apply a filtering mechanism based on the tendency of estimated residuals during fine-tuning. This mitigates the risk of poor performance caused by anomalous samples. We validate the effectiveness and robustness of the proposed HSML method using two datasets containing multiple real-world scenarios. Our experimental results demonstrate that HSML outperforms existing models in terms of identification accuracy, robustness and generalization performance.**

***Index Terms*—Channel impulse response (CIR), cross-scenario and cross-domain, meta-learning, non line-of-sight (NLOS) identification, ultrawideband (UWB) positioning.**

I. Introduction

THE Internet of Things (IoT) integrates various information nodes to enable the seamless connectivity between different objects, and the growing demand for a more convenient lifestyle has further driven the development of high-accuracy indoor positioning in IoT scenarios [1], [2], such as Bluetooth, WiFi, ZigBee, near field communication (NFC), intelligent reflecting surface (IRS), and ultrawideband (UWB)-based positioning [3], [4], [5], [6], [7], [8], [9]. With the widespread deployment of WiFi devices, WiFi positioning has garnered increasing research interest [3], [10], [11]. However, WiFi positioning often faces significant challenges such as signal interference, multipath effects, and fluctuating signal strengths caused by environmental factors [12], [13]. In contrast, UWB offers high-resolution positioning and performs more reliably in environments with significant multipath reflections, while also eliminating the need for extensive data collection [8], [14]. These advantages have led to UWB's broad adoption across various fields, including manufacturing facilities and tunnels. However, in practical implementations, various obstacles significantly obstruct and reflect UWB radio signals, leading to multipath interference and resulting in nonline-of-sight (NLOS) conditions. Compared to LOS conditions, the transmitted signals under NLOS conditions experience varying degrees of time delay, attenuation, and distortion, which significantly degrade UWB positioning accuracy [15], [16], [17]. Therefore, identifying LOS/NLOS propagation conditions and eliminating NLOS errors are crucial issues in realizing high-precision indoor positioning.

Over the years, researchers have been exploring methods for NLOS identification. By virtue of its high spatial and temporal resolution, channel impulse response (CIR) information has become a primary measurement in NLOS identification [18]. Recent advancements in this field include techniques such as feature extraction [19], machine learning (ML) [17], and deep learning (DL) [20]. The main advantage of feature extraction is its easy implementation, but its identification accuracy is greatly affected by the quality of the collected CIR data. When data quality is low, it becomes difficult to extract robust features, which significantly limits identification accuracy. ML and DL methods can better model the nonlinear relationships between input features and labels that represent LOS or NLOS, but they typically require large datasets for training. Besides, these methods almost suffer from a significant drawback: their performance is limited to the specific scenarios in which the training samples are collected. Due to the considerable distribution differences between CIR waveforms from different scenarios or different domains (e.g., device and time domains), the identification accuracy drops sharply when tested on unfamiliar scenarios. As shown in

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Yinong Liu, Haonan Si, Gordon Owusu Boateng, Xiansheng Guo, Yu Cao, and Bocheng Qian are with the Department of Electronic Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: liuyinonghb@163.com; sihaonan@std.uestc.edu.cn; boateng-gordon48@gmail.com; xsguo@uestc.edu.cn; cao_yu@std.uestc.edu.cn; bocheng_qian@std.uestc.edu.cn).

Nirwan Ansari is with the Advanced Networking Laboratory, Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, NJ 07102 USA (e-mail: nirwan.ansari@njit.edu).

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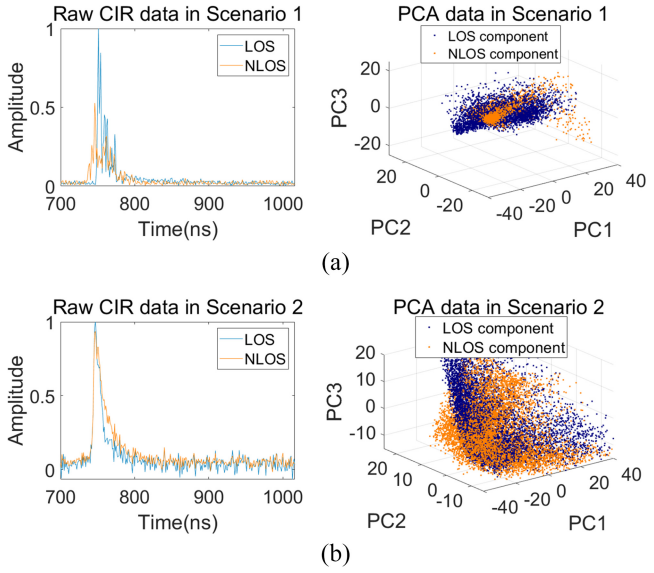


Fig. 1. Comparison of LOS and NLOS CIRs in different scenarios. (a) Raw CIR data and CIR results after PCA processing in Scenario 1. (b) Raw CIR data and CIR results after PCA processing in Scenario 2.

Fig. 1, we conducted a small experiment to illustrates the phenomenon, where CIR samples collected in two different scenarios by different devices. We extract the main features of CIR samples by principal component analysis (PCA) for visualization. We can observe that there are significant cross-scenario and cross-domain differences in LOS and NLOS CIRs, leading to substantial identification errors. Hence, cross-scenario and cross-domain NLOS identification methods merit more research attention from industry and academia.

To enhance the versatility of NLOS identification model in different scenarios and different domains, recent studies have focused on using transfer learning (TL) for NLOS identification [16], [21]. This model typically involves first training the network in a known scenario and then collecting a small number of samples in a new scenario to fine-tune and optimize the network, enabling it to achieve high identification accuracy in the new scenario. However, these TL models are typically only suitable for two specific scenarios, limiting their cross-scenario applicability. Meta-learning, on the other hand, trains a model to quickly adapt to new tasks with just a few labeled samples, achieved by learning from a large number of similar tasks [22], [23], [24]. Hence, this method is well-suited for NLOS identification in multiple scenarios. The method trains a set of generalizable initialization parameters by multiple tasks from different scenarios in the meta-training phase, and fine-tunes the initial parameters with few samples in the meta-testing phase. However, in the meta-training phase, existing meta-learning approaches employ completely random sampling, which cannot fully exploit the potential of samples. Additionally, the quality of randomly sampled data in the meta-testing phase can significantly degrade the performance of fine-tuning of initial parameters. Hence, a modified meta-learning framework is imperative to address these challenges and further improve the robustness and performance of NLOS identification.

In this article, we propose HSML, a hard sample meta-learning method for cross-scenario and cross-domain NLOS identification-based UWB positioning. First, by using CIR samples from multiple scenarios, we train a two-loop learning network for multiple tasks. Then, we screen out the hard samples based on residuals, reconstruct a new dataset, and retrain the network using a residual correction-based focal loss. Subsequently, we fine-tune the neural network (NN) using a small number of samples collected from the target scenario. To ensure the stability of the identification performance, we design a fine-grained meta-testing method based on the magnitudes and variations of the sample residuals. In summary, the main contributions of our work are outlined as follows.

1) We propose a meta-learning-based framework for NLOS identification that is capable of achieving cross-scenario and cross-domain generalization and quickly adapting to a new environment. The proposed model exhibits enhanced NLOS identification accuracy and robustness in a new scenario with only a few samples.
2) Unlike existing meta-learning frameworks [23], [24], which overlook the influence of different samples on the model, particularly those that are to be categorized, we screen samples with extreme residuals and retrain the identification model using a modified loss function. This approach effectively improves generalization across multiple tasks.
3) Moreover, we design a fine-grained meta-testing method based on the magnitudes and variations of sample residuals. This method is capable of screening out anomalous samples, ensuring stable NLOS identification in the target environment.
4) In addition to validating the performance of our proposed method on a dataset collected from multiple real-world scenarios, we also test our method on another dataset that exhibits significant differences across several dimensions. This new dataset was collected by a different set of devices and processed by different data collection parameters and data preprocessing methods. Experimental results and analysis prove that our proposed method consistently achieves high NLOS identification accuracy, validating its outstanding robustness against cross-scenario and cross-domain tasks.

The remainder of this article is organized as follows. Section II introduces related work, and Section III presents the base formulation of the NLOS problem and meta-learning. Then, the proposed HSML is presented in detail in Section IV. Section V discusses the results of testing on publicly available datasets and compares our model with state-of-the-art NLOS identification methods. Finally, Section VI concludes this article.

Notations: a, $\boldsymbol{a}$, $\boldsymbol{A}$, and $\mathcal{A}$ stand for a scalar, a column vector, a set, and a task, respectively. $|a|$ represents the absolute value of scalar a, and $|\mathcal{A}|$ denotes the cardinality of set $\mathcal{A}$.

II. Related Works

In this section, we briefly overview existing works on NLOS identification, cross-scenario NLOS identification, and meta-learning approaches.

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A. NLOS Identification

Fundamentally, NLOS Identification methods can be categorized into feature extraction [17], [19], [25], ML [26], and DL-based methods [20], [27]. Yang et al. [17] proposed a method for NLOS identification that involves calculating six physical features derived from both raw CIR data and corresponding chip output parameters. These features are then used exclusively with a support vector machine (SVM) for classification. Ferreira et al. [19] combined various real-time features with different ML algorithms, leveraging UWB transceivers to identify and mitigate NLOS conditions.

Generally speaking, ML [26] and DL-based methods [20], [27] are more commonly utilized for NLOS identification due to their powerful nonlinear approximation capabilities. For ML approaches, Cui et al. [26] proposed an LOS/NLOS identification method based on Morlet wavelet transform and convolutional NNs (MWT-CNN) to enhance NLOS identification capability in static scenarios. For DL approaches, Pei et al. [27] employed a fully convolutional network and attention (FCN-Attention)-based method to achieve an accuracy of 88.24% when performing UWB NLOS identification using complete CIR data. In addition, Wu et al. [20] applied a self-attention mechanism to resource-constrained edge devices and employed a data-driven feature selection method to enhance model performance, significantly improving NLOS identification on these devices.

The above approaches and other related works for NLOS identification have achieved remarkable results. However, the main limitations of these methods are as follows.

1) *Performance in Specific Scenarios:* The effectiveness of the aforementioned methods is limited by the specific scenarios in which they are applied. In unknown or dynamic scenarios, their performance significantly reduces due to changes in the distribution of CIRs. Therefore, these methods require training a new identification model from scratch for each scenario, which is time-consuming and often impractical for most real-world applications.
2) *Ambient Noise Difference:* Each identification model learns specific noise originating from a given scenario. Nonetheless, the distribution of noise varies in different scenarios. As a result, the performance of the well-trained model will degrade significantly when applied in a new scenario.

Due to these shortcomings, recent research has increasingly focused on enhancing model robustness and realizing accurate identification in multiple scenarios.

B. NLOS Identification in Multiple Scenarios

Currently, mainstream methods enabling robust NLOS identification in multiple scenarios are based on TL [16], [21]. Fontaine et al. [21] initially trained a convolutional NN (CNN) model using CIR samples from a source scenario, and then fine-tuned the model with data from the target scenario. Nkrow et al. [16] proposed a robust TL-based NLOS identification approach, which constructs representative homogeneous features for source and target scenarios via cross-domain mappings.

These methods provide ideas for enhancing the performance of the model in a new scenario, but they still suffer from the following shortcomings.

1) *Poor Cross-Scenario Capabilities:* Most TL-based NLOS identification methods are only suitable for two specific scenarios. Insufficient knowledge acquired from the source domain limits the generalization ability of the model. Consequently, substantial additional effort is required to adapt and fine-tune the model for new scenarios.
2) *Lack of Cross-Domain Capabilities:* These studies are limited to CIR samples collected by the same devices. When applied to different datasets collected using different devices, their performance degrades significantly. Therefore, their above methods lack cross-domain (device domain) generalization capabilities.

C. Meta-Learning Approach

Meta-learning methods [24] can be categorized into metric learning [28], [29], memory architectures [22], [30], [31], and gradient-based approaches [23]. Metric learning involves creating a similarity space where learning with a small number of samples becomes particularly effective. Ye et al. [28] proposed a novel approach to adapt the instance embeddings to the target classification task using a set-to-set function, resulting in task-specific and highly discriminative embeddings. Dvornik et al. [30] developed an ensemble method for few-shot classification by training multiple deep networks, using pairwise penalties for cooperation and diversity, and then applying distillation to create a single network. Gradient descent is employed to adjust the optimization algorithm so that the model can converge within a small number of optimization steps. Finn et al. [23] proposed a meta-learner called model-agnostic meta-learning (MAML), which learns how to effectively initialize a base learner to complete a new task.

Among the meta learning methods, MAML is a relatively general optimization algorithm that is compatible with any model within the framework of gradient descent, which optimizes the meta-parameters by using the validation loss of the base learner through gradient descent. By virtue of its outstanding cross-task generalization capabilities, MAML is suitable for cross-scenario and cross-domain learning tasks. Notably, MAML selects samples in a completely random manner during meta-training, overlooking the influence of different samples on the model. However, in few-shot learning applications, individual samples, especially the hard ones, can significantly affect the learning performance of MAML. To improve the learning efficiency and generalization capabilities, there has been increasing focus on hard sample mining techniques.

D. Hard Sample Mining

Hard sample mining was proposed by Shrivastava et al. [32] for target detection using a deep NN (DNN). It treats image

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proposals that overlap with the ground truth and cause more confusion as hard negative samples. Training on these more cluttered data enhances the robustness and performance of the detection model [22], [33], [34]. Sun et al. [22] tried to collect more difficult tasks online so that the meta-TL (MTL) model could "grow faster and stronger through more hardness."

Furthermore, Zhou and Wu [35] comprehensively discussed the methods for classifying hard samples and the approach for handing them. Most existing measures can be divided into four categories: 1) prediction-based measures [36], [37]; 2) category proportion-based measures [38]; 3) margin-based measures [39]; and 4) uncertainty-based measures [40]. Category proportion-based measures are typically used in multicategory classification scenarios with highly imbalanced categories, and margin-based measures are more suitable for situations with strong geometric spatial interpretability. Uncertainty-based measures, on the other hand, are applicable to multicategory classification problems or tasks that require evaluating the model's confidence. Since NLOS identification is neither a multicategory classification problem nor one involving strong geometric spatial interpretability, the aforementioned methods are not applicable. Prediction-based measures focus primarily on the samples themselves and do not rely on the specific characteristics of the model. With a clear task objective, these measures are intuitive and easy to implement, making them well-suited to our problem. Therefore, we adopt this approach to identify difficult samples.

In summary, most existing studies focus on model performance in a single static scenario, lacking cross-scenario and cross-domain generalization capabilities. To the best of our knowledge, this study is the first to propose an HSML approach specifically designed to address the challenges of NLOS identification. Our approach enables rapid adaptation of the identification model to new environments, even when only a limited number of samples are available. This represents a significant advancement in improving model performance and robustness in diverse and dynamically changing scenarios.

III. Problem Formulation

A. Basic Parameter Description

Assume that there are a total of P different scenarios for training NN parameters. The ith scenario has Q_i tags (UWB signal transmitting equipment) and C_i base stations (UWB signal receiving equipment), where $i = 1, 2, \ldots, P$. In the ith scenario, tag q_i consistently sends signals to the surrounding base station c_i, where $q_i = 1, 2, \ldots, Q_i$ and $c_i = 1, 2, \ldots, C_i$. If there are no blockages or obstructions between tag q_i and base station c_i, the condition is defined as LOS; otherwise, it is considered NLOS.

For simplicity, we consider using the same set of labels and base stations in both source and target domains, and the set contains only one label and one base station. We denote the ith source domain scenario as $\boldsymbol{X}_{\mathcal{S}_i}$ and the entire source domain as $\boldsymbol{X}_{\mathcal{S}} = \{\boldsymbol{X}_{\mathcal{S}_i}\}_{i=1}^{P}$. The target domain scenario is represented as $\boldsymbol{X}_{\mathcal{T}}$. In $\boldsymbol{X}_{\mathcal{S}_i}$, we collect $|\boldsymbol{X}_{\mathcal{S}_i}|$ CIR samples, expressed as

$$\boldsymbol{X}_{\mathcal{S}_i} = \left\{\boldsymbol{X}^l_{\mathcal{S}_i}, \boldsymbol{X}^n_{\mathcal{S}_i}\right\} \in \mathbb{R}^{|\boldsymbol{X}_{\mathcal{S}_i}| \times d} \tag{1}$$

where $\boldsymbol{X}^l_{\mathcal{S}_i} = \{\boldsymbol{x}^{l,j}_{\mathcal{S}_i}\}_{j=1}^{|\boldsymbol{X}^l_{\mathcal{S}_i}|}$ and $\boldsymbol{X}^n_{\mathcal{S}_i} = \{\boldsymbol{x}^{n,k}_{\mathcal{S}_i}\}_{k=1}^{|\boldsymbol{X}^n_{\mathcal{S}_i}|}$ represent the LOS and NLOS datasets in the ith source domain scenario, respectively. Superscripts l and n represent LOS and NLOS components. $\boldsymbol{x}^{l,j}_{\mathcal{S}_i}$ and $\boldsymbol{x}^{n,k}_{\mathcal{S}_i}$ represent the jth LOS and the kth NLOS CIR samples in the ith source domain, respectively. We have $|\boldsymbol{X}_{\mathcal{S}_i}| = |\boldsymbol{X}^l_{\mathcal{S}_i}| + |\boldsymbol{X}^n_{\mathcal{S}_i}|$. d representing the length of a CIR sample and $\boldsymbol{y}_{\mathcal{S}_i} = \{\boldsymbol{y}^l_{\mathcal{S}_i}, \boldsymbol{y}^n_{\mathcal{S}_i}\}$ representing the corresponding labels of $\boldsymbol{X}_{\mathcal{S}_i}$. $\boldsymbol{y}^l_{\mathcal{S}_i} = \{y^{l,j}_{\mathcal{S}_i}\}_{j=1}^{|\boldsymbol{X}^l_{\mathcal{S}_i}|}$ represents the LOS label from the ith source domain scenario, and $\boldsymbol{y}^n_{\mathcal{S}_i} = \{y^{n,k}_{\mathcal{S}_i}\}_{k=1}^{|\boldsymbol{X}^n_{\mathcal{S}_i}|}$ represents the NLOS label from the ith source domain scenario.

In the target domain, we collect few samples to fine-tune the NN parameters. The data in the target domain is expressed as

$$\boldsymbol{X}_{\mathcal{T}} = \left\{\boldsymbol{X}^l_{\mathcal{T}}, \boldsymbol{X}^n_{\mathcal{T}}\right\} \in \mathbb{R}^{|\boldsymbol{X}_{\mathcal{T}}|} \tag{2}$$

where $\boldsymbol{X}^l_{\mathcal{T}} = \{\boldsymbol{x}^{l,p}_{\mathcal{T}}\}_{p=1}^{|\boldsymbol{X}^l_{\mathcal{T}}|}$ and $\boldsymbol{X}^n_{\mathcal{T}} = \{\boldsymbol{x}^{n,q}_{\mathcal{T}}\}_{q=1}^{|\boldsymbol{X}^n_{\mathcal{T}}|}$ represent the LOS and NLOS datasets in the target domain scenario, respectively. $\boldsymbol{x}^{l,j}_{\mathcal{T}}$ and $\boldsymbol{x}^{n,k}_{\mathcal{T}}$ represent the jth LOS and kth NLOS CIR samples of the target domain, respectively. The corresponding labels of $\boldsymbol{X}_{\mathcal{T}}$ are represented as $\boldsymbol{y}_{\mathcal{T}} = \{\boldsymbol{y}^l_{\mathcal{T}}, \boldsymbol{y}^n_{\mathcal{T}}\}$, where $\boldsymbol{y}^l_{\mathcal{T}} = \{y^{l,p}_{\mathcal{T}}\}_{p=1}^{|\boldsymbol{X}^l_{\mathcal{T}}|}$ and $\boldsymbol{y}^n_{\mathcal{T}} = \{y^{n,q}_{\mathcal{T}}\}_{q=1}^{|\boldsymbol{X}^n_{\mathcal{T}}|}$. $y^{l,p}_{\mathcal{T}}$ represents the pth label in $\boldsymbol{y}^l_{\mathcal{T}}$ and $y^{n,q}_{\mathcal{T}}$ represents the qth label in $\boldsymbol{y}^n_{\mathcal{T}}$. For notation conciseness, we label the CIR samples of LOS and NLOS as 0 and 1, respectively. That is, $y^{l,j}_{\mathcal{S}_i} = y^{l,p}_{\mathcal{T}} = 0$ and $y^{n,k}_{\mathcal{S}_i} = y^{n,q}_{\mathcal{T}} = 1$ for $i = 1, 2, \ldots, P$.

B. Meta-Learning Preliminaries

Consider a model, denoted by $f_{\boldsymbol{\theta}}$ with corresponding parameters $\boldsymbol{\theta}$, which maps input measurement $\boldsymbol{X}$ to output y. The parameters $\boldsymbol{\theta}$ contain $\boldsymbol{\theta}_{\text{out}}$ and $\boldsymbol{\theta}_i$, $i = 1, 2, \ldots, P$, which represent the parameters of the outer loop and inner loop, respectively. In each iteration, the inner loop learns the specific features of each task, while the outer loop captures cross-task generalization properties using a cross-entropy loss. Within the meta-learning framework, tasks are drawn from the distribution $p(\mathcal{M})$, denoted as $\mathcal{M}_i \sim p(\mathcal{M})$. Then, it considers a distribution over tasks $p(\mathcal{M})$, where each task includes a support set for training and a query set for testing.

In the meta-training phase, P training tasks are sampled from a specific distribution $p(\mathcal{M})$ and the corresponding datasets $\boldsymbol{X}_{\mathcal{S}}$ are transmitted to the model. In the meta-testing phase, a new test task $\mathcal{M}_{\mathcal{T}}$ is presented, consisting of a small support set and a query set. The new task $\mathcal{M}_{\mathcal{T}}$ also follows the distribution $p(\mathcal{M})$. The objective of meta-learning is to learn a set of initial parameters $\boldsymbol{\theta}^*$ that can quickly adapt to the new test task by using the small support set with only a few gradient descent steps and perform well on the query set.

At the beginning of each iteration, we perform $\boldsymbol{\theta}_{\text{out}} = \boldsymbol{\theta}_i$. In the inner loop, the support set is used to update the parameters $\boldsymbol{\theta}_i$, expressed as

$$\boldsymbol{\theta}'_i = \boldsymbol{\theta}_i - \alpha \nabla_{\boldsymbol{\theta}_i} \mathcal{L}_{\mathcal{M}_i}\left(f_{\boldsymbol{\theta}_i}\right) \tag{3}$$

where $\boldsymbol{\theta}'_i$ represents the result of $\boldsymbol{\theta}_i$ after iterations on the support set, α denotes the step size of the inner loop, and $\mathcal{L}_{\mathcal{M}_i}(f_{\boldsymbol{\theta}_i})$ denotes the loss value which is computed based on

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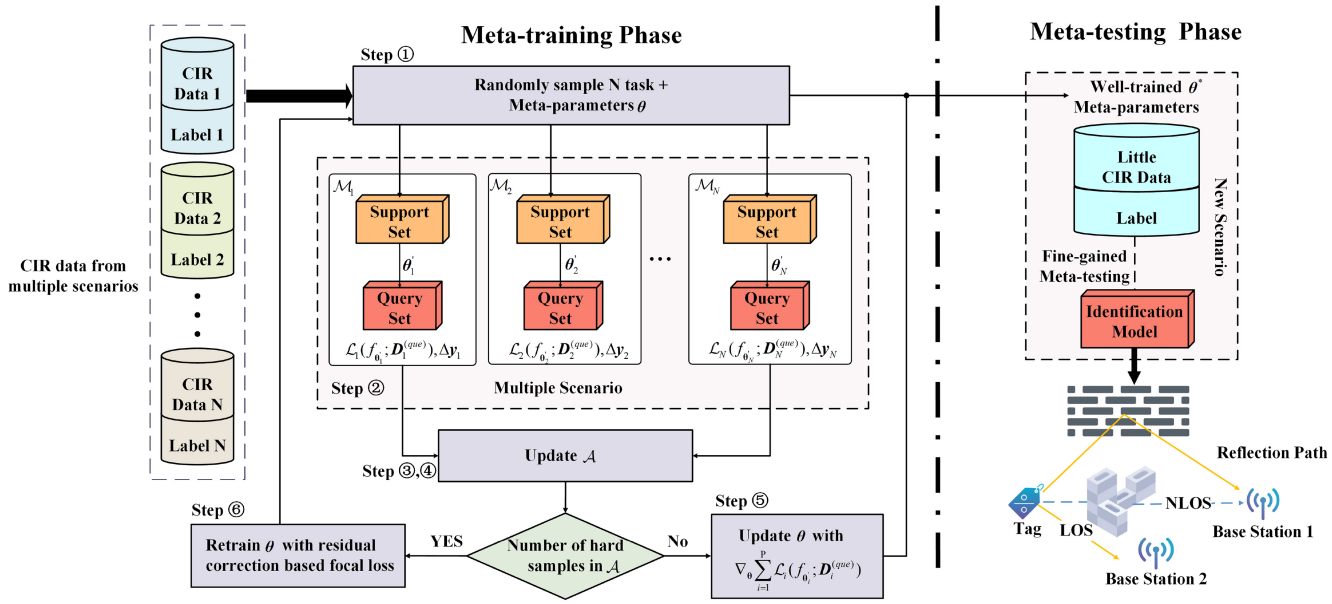


Fig. 2. Overview of HSML.

the support set. Then, $\boldsymbol{\theta}_i'$ is used to obtain the loss value of the ith task on the query set. The outer loop formulates a meta-optimization problem to find $\boldsymbol{\theta}_{\text{out}}$ as

$$\min_{\boldsymbol{\theta}} \sum_{\mathcal{M}_i \sim p(\mathcal{M})} \mathcal{L}_{\mathcal{M}_i}\left(f_{\boldsymbol{\theta}_i'}\right) = \sum_{\mathcal{M}_i \sim p(\mathcal{M})} \mathcal{L}_{\mathcal{M}_i}\left(f_{\boldsymbol{\theta}_i - \alpha \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{M}_i}(f_{\boldsymbol{\theta}_i})}\right). \tag{4}$$

Next, the meta-parameters are updated via stochastic gradient descent (SGD) as

$$\boldsymbol{\theta}_{\text{out}} \longleftarrow \boldsymbol{\theta}_{\text{out}} - \beta \nabla_{\boldsymbol{\theta}} \sum_{\mathcal{M}_i \sim p(\mathcal{M})} \mathcal{L}_{M_i}\left(f_{\boldsymbol{\theta}_i'}\right) \tag{5}$$

where β denotes the step size of the outer loop and $\mathcal{L}_{M_i}(f_{\boldsymbol{\theta}_i'})$ computes the loss on the query set. By repeating the aforementioned steps in (3)–(5), the initialized meta-parameters $\boldsymbol{\theta}^*$ after convergence are obtained.

During the meta-testing phase, we divide the target task $\mathcal{M}_{\mathcal{T}}$ into a support set containing a few samples and a query set comprising the remaining samples. The parameters used for testing on the query set of the target task are denoted as $\boldsymbol{\theta}_{\mathcal{M}_{\mathcal{T}}}$. To achieve high identification accuracy on the query set, we initialize $\boldsymbol{\theta}_{\mathcal{M}_{\mathcal{T}}}$ with the pretrained meta-parameters $\boldsymbol{\theta}^*$ and fine-tune $\boldsymbol{\theta}_{\mathcal{M}_{\mathcal{T}}}$ using the support set. Then, the fine-tuning process can be formulated as follows:

$$\boldsymbol{\theta}_{\mathcal{M}_{\mathcal{T}}} \leftarrow \boldsymbol{\theta}_{\mathcal{M}_{\mathcal{T}}} - \alpha \nabla_{\boldsymbol{\theta}_{\mathcal{M}_{\mathcal{T}}}} \mathcal{L}_{\mathcal{M}_{\mathcal{T}}}\left(\boldsymbol{\theta}_{\mathcal{M}_{\mathcal{T}}}\right) \tag{6}$$

where $\mathcal{L}_{\mathcal{M}_{\mathcal{T}}}(\boldsymbol{\theta}_{\mathcal{M}_{\mathcal{T}}})$ denotes the loss function on the support set.

IV. Proposed Algorithm

As shown in Fig. 2, we propose an HSML framework based on meta-learning, which consists of two stages to achieve cross-domain and cross-scenario UWB NLOS identification, i.e., hard sample meta-training and fine-grained meta-testing.

A. Hard Sample Meta-Training

Step ①: The NN is initialized with a preselected network architecture and a set of randomly initialized parameters $\boldsymbol{\theta}$, which are represented by the function $f_{\boldsymbol{\theta}}$. These parameters serve as the initial values for outer parameters $\boldsymbol{\theta}_{\text{out}}$ and inner parameters $\boldsymbol{\theta}_i$ during the first iteration. A total of P tasks, represented as $\mathcal{M}_1, \ldots, \mathcal{M}_P$, are separately sampled from the source domain datasets $\boldsymbol{X}_{\mathcal{S}_i}$ to be used as the training tasks ($i = 1, 2, \ldots, P$). Each task represents a different scenario and follows the identification task distribution $p(\mathcal{M})$. Hence, the cross-scenario and cross-domain NLOS identification is transformed into a multitask learning problem. The ith task $\mathcal{M}_i = \{\mathcal{L}_i; \boldsymbol{D}_i^{(\text{sup})}, \boldsymbol{D}_i^{(\text{que})}\}$ is composed of a loss function $\mathcal{L}_i$, a support set $\boldsymbol{D}_i^{(\text{sup})}$ and a query set $\boldsymbol{D}_i^{(\text{que})}$, where $\boldsymbol{D}_i^{(\text{sup})} = \{\boldsymbol{u}_{i,j}^{(\text{sup})}, y_{i,j}^{(\text{sup})}\}_{j=1}^{|\boldsymbol{D}_i^{(\text{sup})}|}$, $\boldsymbol{D}_i^{(\text{que})} = \{\boldsymbol{u}_{i,k}^{(\text{que})}, y_{i,k}^{(\text{que})}\}_{k=1}^{|\boldsymbol{D}_i^{(\text{que})}|}$. $\boldsymbol{u}_{i,j}^{(\text{sup})}$ represents the jth sample in $\boldsymbol{D}_i^{(\text{sup})}$ and $y_{i,k}^{(\text{sup})}$ is the corresponding label. $\boldsymbol{u}_{i,k}^{(\text{que})}$ represents the kth sample in $\boldsymbol{D}_i^{(\text{que})}$ and $y_{i,k}^{(\text{que})}$ is the corresponding label. For binary identification problems, the following cross-entropy loss function is generally used:

$$\mathcal{L}_i\left(f_{\boldsymbol{\theta}_i}; \boldsymbol{D}_i^{(\text{sup})}\right) = \sum_{j=1}^{|\boldsymbol{D}_i^{(\text{sup})}|} \left[y_{i,j}^{(\text{sup})} \log f_{\boldsymbol{\theta}_i}\left(\boldsymbol{u}_{i,j}^{(\text{sup})}\right) + \left(1 - y_{i,j}^{(\text{sup})}\right) \log\left(1 - f_{\boldsymbol{\theta}_i}\left(\boldsymbol{u}_{i,j}^{(\text{sup})}\right)\right)\right] \tag{7}$$

where $f_{\boldsymbol{\theta}_i}(\boldsymbol{u}_{i,j}^{(\text{sup})})$ represents the predicted output of $\boldsymbol{u}_{i,j}^{(\text{sup})}$.

Step ②: For the identification task $\mathcal{M}_i$, we train $\boldsymbol{\theta}_i$ with the support set $\boldsymbol{D}_i^{(\text{sup})}$ and calculate the task-specific loss $\mathcal{L}_i(f_{\boldsymbol{\theta}_i}; \boldsymbol{D}_i^{(\text{sup})})$ in (7). We obtain task-specific parameters $\boldsymbol{\theta}_i'$ during the inner loop using one-step gradient descent update in (3). The task-specific parameter $\boldsymbol{\theta}_i'$ provides only limited information about each positioning task, as they are derived using an one-step gradient descent update on the support set $\boldsymbol{D}_i^{(\text{sup})}$. Further evaluation of the performance on the

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query set $\boldsymbol{D}_i^{(\text{que})}$ is necessary to obtain a more comprehensive understanding of each task.

Step ③: In this step, we employ prediction-based measures to mitigate the impact of hard samples, as analyzed in Section II-D. The core idea behind this is that a higher loss (or a lower predicted probability) for the ground truth signifies greater difficulty. Drawing inspiration from [35] and in line with the various loss functions discussed in this article, we adopt the difference between predicted results and true labels as the criterion for identifying difficult samples. To facilitate this process, we initialize a new training dataset, $\mathcal{A}$, with $\mathcal{A} = \varnothing$. Subsequently, we calculate the residual of each sample, laying the groundwork for hard sample meta-training. For the pth sample $\boldsymbol{u}_{i,p}^{(\text{que})}$, we can obtain the corresponding label $y_{i,p}^{(\text{que})}$ and the prediction result $\hat{y}_{i,p}^{(\text{que})}$, from which we can calculate the residual $\Delta y_{i,p}$ as

$$\Delta y_{i,p} = \left| \hat{y}_{i,p}^{(\text{que})} - y_{i,p}^{(\text{que})} \right|. \tag{8}$$

If $\Delta y_{i,p} > \gamma$, that is, the predicted result significantly differs from the actual label, we classify this sample as a hard sample. Conversely, if $\Delta y_{i,p} < \mu$, that is, the predicted result is close to the actual label, we classify it as soft sample soft sample. It should be noted that $0 < \mu < \gamma < 1$, e.g., if $\gamma = 0.7$ and $\mu = 0.25$, then we store $\boldsymbol{u}_{i,p}^{(\text{que})}$ in $\mathcal{A}$ if $\Delta y_{i,p} \in [0, 0.25]$ or $\Delta y_{i,p} \in [0.7, 1]$. These two cases are both extreme cases of residuals, and their corresponding samples are both stored in the new dataset $\mathcal{A}$. Hence, we can define our sample selection method using the following formula:

$$\mathcal{A} \leftarrow \begin{cases} \mathcal{A} \cup \left\{ \boldsymbol{u}_{i,p}^{(\text{que})} \right\}, & \Delta y_{i,p} > \gamma \\ \mathcal{A} \cup \left\{ \boldsymbol{u}_{i,p}^{(\text{que})} \right\}, & \Delta y_{i,p} < \mu, \\ \mathcal{A}, & \text{otherwise} \end{cases} \tag{9}$$

If $|\mathcal{A}|$ does not reach the predefined number of hard samples threshold m, the number of hard samples is relatively small. In this case, we execute step ④ and step ⑤; otherwise, we execute step ⑥.

Step ④: $\mathcal{L}_i(f_{\boldsymbol{\theta}_i'}; \boldsymbol{D}_i^{(\text{que})})$ can be calculated in *Step* ③ to further evaluate the performance of one-step update for the ith identification task, using the samples in the query set, while simultaneously handling difficult samples. After that, we update the parameters $\boldsymbol{\theta}_{\text{out}}$ by minimizing the sum of all task-specific losses. In this way, the meta-optimization across all P tasks is given by

$$\boldsymbol{\theta}_{\text{out}}' = \arg\min_{\boldsymbol{\theta}_{\text{out}}'} \sum_{i=1}^{P} \mathcal{L}_i\left(f_{\boldsymbol{\theta}_i'}; \boldsymbol{D}_i^{(\text{que})}\right). \tag{10}$$

It is important to note that during the meta-optimization process, optimization is performed on the parameters $\boldsymbol{\theta}_{\text{out}}$, while $\boldsymbol{\theta}_i'$ is only used to calculate the task-specific loss $\mathcal{L}_i(f_{\boldsymbol{\theta}_i'}; \boldsymbol{D}_i^{(\text{que})})$. This mechanism ensures that the meta-parameters $\boldsymbol{\theta}_{\text{out}}$ are updated to optimize generalization performance across all tasks, while the task-specific parameters $\boldsymbol{\theta}_i$ are used to assess the one-step update performance for each individual task.

Step ⑤: The meta-optimization of the outer loop is conducted using Adam optimizer [41], which is accomplished through the following iterative steps.

1) *Compute Gradients:* We first compute the gradients as follows:
$$\nabla J(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \sum_{i=1}^{P} \mathcal{L}_i\left(f_{\boldsymbol{\theta}_i'}; \boldsymbol{D}_i^{(\text{que})}\right) \tag{11}$$
2) *Update the First Moment Estimate:* Next, we update the first moment estimate as:
$$\varpi_t = \beta_1 \varpi_{t-1} + (1 - \beta_1) \nabla J(\boldsymbol{\theta}) \tag{12}$$
if $t = 1$, $\varpi_0 = 0$ and β_1 is a constant close to 1 (e.g., 0.9).
3) *Update the Second-Moment Estimate:* Then, we update the second-moment estimate as:
$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)(\nabla J(\boldsymbol{\theta}))^2 \tag{13}$$
if $t = 1$, $v_0 = 0$ and β_2 is also a constant close to 1 (e.g., 0.99).
4) *Deviation Correction:* After that, we apply deviation correction as:
$$\hat{\varpi}_t = \frac{\varpi_t}{1 - (\beta_1)^t} \tag{14}$$
$$\hat{v}_t = \frac{v_t}{1 - (\beta_2)^t} \tag{15}$$
5) *Update Parameters:* Finally, we update the parameters as:
$$\boldsymbol{\theta}_{\text{out}}' = \boldsymbol{\theta}_{\text{out}} - \eta \frac{\hat{\varpi}_t}{\sqrt{\hat{v}_t} + \epsilon} \tag{16}$$
where η represents the step size, and ϵ is a very small value to prevent division by zero errors. The variable t in all the formulas of the Adam optimizer represents the tth iteration and starts counting from 1.

Adam computes adaptive learning rates for each parameter, making optimization more stable and efficient [41]. The first-moment estimate, ϖ_t, smoothens noisy gradients, reducing oscillations and accelerating convergence, especially in regions with small or fluctuating gradients. The second-moment estimate, v_t, adjusts the learning rate based on gradient variance, preventing overly large or small updates and further stabilizing training. Bias correction terms, $\hat{\varpi}_t$ and $\hat{v}_t$, address initialization bias, particularly significant in the early stages. This adaptability enables Adam to reduce the need for manual learning rate tuning, improve stability in noisy or sparse gradients, and achieve faster convergence in practice.

Step ⑥: To enhance the model robustness in the existence of hard samples, we use $\mathcal{A}$ to retrain the identification model in this step, as depicted in Fig. 3.

If the number of hard samples reaches the threshold, we stop the meta-training process, load the current NN parameters $\boldsymbol{\theta}_{\text{out}}$, and the dataset $\mathcal{A}$, where $\mathcal{A} = \{\boldsymbol{u}_v, y_v\}_{v=1}^{|\mathcal{A}|}$, $\boldsymbol{u}_v$ represents the vth sample in $\mathcal{A}$ and y_v is the corresponding label. To further improve model performance, we design a residual correction-based focal loss, formulated as

$$\text{FocalLoss}_v = \left(1 - \frac{1}{e^{\mathcal{L}_v}}\right)^{\lambda} \mathcal{L}_v \tag{17}$$

with $\mathcal{L}_v$ being the cross entropy loss value of the vth sample, given by

$$\mathcal{L}_v = y_v \log f_{\boldsymbol{\theta}_{\text{out}}}(\boldsymbol{u}_v) + (1 - y_v) \log\left(1 - f_{\boldsymbol{\theta}_{\text{out}}}(\boldsymbol{u}_v)\right). \tag{18}$$

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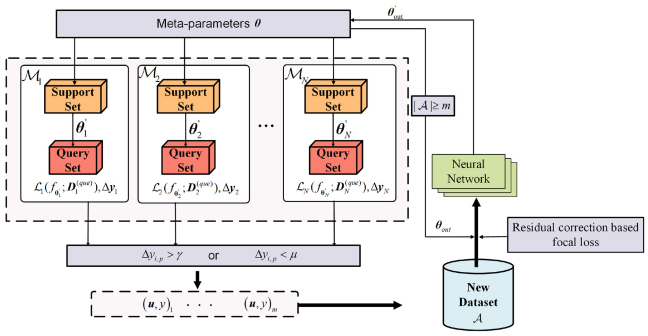


Fig. 3. Hard sample component in our proposed framework.

Then, we define the residual correction-based focal loss function as

$$\mathcal{L}^*\left(f_{\boldsymbol{\theta}_{\text{out}}}, \mathcal{A}\right) = \sum_{v=1}^{|\mathcal{A}|} \mathcal{L}^*\left(f_{\boldsymbol{\theta}_{\text{out}}}, \{\boldsymbol{u}_v, y_v\}\right) \tag{19}$$

with

$$\mathcal{L}^*\left(f_{\boldsymbol{\theta}_{\text{out}}}, \{\boldsymbol{u}_v, y_v\}\right) = \begin{cases} \text{FocalLoss}_v, & \Delta y_v \geq \zeta \\ \text{FocalLoss}_v \times \tau, & \Delta y_v < \zeta \end{cases} \tag{20}$$

where Δy_v represents the residual of the vth sample. λ is a focusing factor that regulates the attention given to hard samples. Typically, a higher λ value encourages the model to focus more on the samples that are harder to identify. ζ represents the threshold for residual correction, which is empirically chosen as a small positive constant (e.g., 0.1). $0 < \tau < 1$ is a hyperparameter of residual correction. Therefore, the proposed loss function can dynamically adjust weights for hard and soft samples, improving accuracy for both.

By combining the residual correction-based focal loss functions, the meta-optimization function can be expressed as

$$\boldsymbol{\theta}'_{\text{out}} = \arg\min_{\boldsymbol{\theta}'_{\text{out}}} \sum_{v=1}^{|\mathcal{A}|} \mathcal{L}^*\left(f_{\boldsymbol{\theta}_{\text{out}}}, \{\boldsymbol{u}_v, y_v\}\right). \tag{21}$$

To achieve this meta-optimization process, we obtain a set of outer-loop parameters by iterating K times using Adam (11)–(16). These parameters are then used as the initial parameters for meta-training for the next iteration.

We then repeat steps ① to ⑥ until the outer-loop parameters $\boldsymbol{\theta}_{\text{out}}$ converge. Finally, we obtain the meta-parameters $\boldsymbol{\theta}^*$. The complete hard sample meta-training algorithm is shown as Algorithm 1.

Algorithm 1 Hard Sample Meta-Training

Require: Task distribution $p(\mathcal{M})$ and corresponding dataset $\boldsymbol{X}_{\mathcal{S}}$, preset hyper-parameters: $\alpha, \beta, \gamma, \mu, \lambda, \tau, \zeta, m, K$;
Ensure: Identification model $f(\cdot)$.

```
Randomly initialize θ;
while not done do
    Randomly sample batch of tasks M_i from X_{S_i};
    for all M_i do
        Divide M_i into D_i^(sup) and D_i^(que);
        Evaluate ∇L_i(f_{θ_i}; D_i^(sup)) and update network parameters in Eq. (3);
        Calculate L_i(f_{θ'_i}; D_i^(que)) and Δy_i in Eqs. (7) and (8);
        Update dataset A based on Eq. (9);
        Calculate the total number of hard samples in A;
        if |A| < m then
            Update θ_out from Eqs. (11)-(16);
        else if |A| ≥ m then
            Break meta-training;
            Iterate K times on A to obtain θ'_out in Eqs. (17) and (19);
            Reinitialize meta-training with θ'_out as the initialization parameter.
        end if
    end for
end while
```

B. Fine-Grained Meta-Testing

In the meta-testing phase, traditional meta-learning methods fine-tune the model using a small number of samples. However, in few-shot learning scenarios, individual samples—particularly anomalous ones—can significantly impact model accuracy. To mitigate the risk of poor performance caused by anomalous samples in the query set after fine-tuning, we propose a fine-grained meta-testing approach.

After the meta-training phase, we obtain the well-trained meta-parameters $\boldsymbol{\theta}^*$, which capture the essential features of NLOS identification across multiple scenarios. Consider an unseen task in the new scenario $\mathcal{M}_{\mathcal{T}} \sim p(\mathcal{M})$ with support set $\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}$ and query set $\boldsymbol{D}_{\mathcal{T}}^{(\text{que})}$. Support set $\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}$ is constructed from a small number of labeled samples selected from $\boldsymbol{X}_{\mathcal{T}}$, where $\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})} = \{\boldsymbol{u}_q, y_q\}_{q=1}^{|\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}|}$. Query set $\boldsymbol{D}_{\mathcal{T}}^{(\text{que})}$ is constructed from the remaining samples in $\boldsymbol{X}_{\mathcal{T}}$, where $\boldsymbol{u}_q$ is the qth sample in $\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}$, and y_q is the corresponding label. We use the pretrained network $f_{\boldsymbol{\theta}^*}$ initialized with a predefined architecture and learned meta-parameters $\boldsymbol{\theta}^*$ as initialization. Then, we update the parameters $\boldsymbol{\theta}^*$ by taking R-step iterations in the meta-training phase.

During the R iterations, residual Δy_1 is calculated after the first iteration, and the residual Δy_c is calculated after the cth iteration ($1 < c < R$). For the qth sample ($1 \leq q \leq t$), if its residual is less than the predefined threshold σ, it is determined that the NN has a good identification effect on the sample. Otherwise, we calculate the difference between the residuals obtained in the first iteration and the cth iteration as

$$\Delta y_{1c,q} = \Delta y_{1,q} - \Delta y_{c,q}. \tag{22}$$

Subsequently, samples can be categorized into either anomalous or clean samples. We define clean samples as those with smaller residuals or whose residuals decrease rapidly during the iterations (i.e., $\Delta y_{1c,q} > b$ or $\Delta y_{c,q} < \sigma$). All other samples are considered anomalous and discarded to reduce the risk of poor performance. b is the predefined threshold that represents the tendency of the residuals from the first iteration to the cth iteration. Hence, the screening of anomalous samples

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Algorithm 2 Fine-Grained Meta-Testing

Require: Task distribution $p(\mathcal{M})$ and corresponding dataset $\boldsymbol{X}_{\mathcal{T}}$, fine-grained meta-testing threshold: σ, b;
Ensure: Prediction result y.
1: Load NN parameters $\boldsymbol{\theta}^*$ and sample $|\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}|$ data-points $\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}$ from $\boldsymbol{X}_{\mathcal{T}}$;
2: **for** $q \in 1, ..., R$ **do**
3: Update the NN parameters $\boldsymbol{\theta}^*$ with $\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}$;
4: **if** $q = 1$ **then**
5: Calculate Δy_1;
6: **else if** $q = c$ **then**
7: Calculate Δy_c and Δy_{1c};
8: Filter out samples in Eq. (23);
9: **end if**
10: **end for**
11: Obtain task-specific optimal parameters $\boldsymbol{\theta}_{\mathcal{T}}^*$;
12: Predict the output of $\boldsymbol{D}_{\mathcal{T}}^{(\text{que})}$ using the parameter $\boldsymbol{\theta}_{\mathcal{T}}^*$.

is expressed as

$$\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})} \leftarrow \begin{cases} \boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}, & \Delta y_{c,q} < \sigma \\ \boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}, & \Delta y_{1c,q} > b \\ \boldsymbol{D}_{\mathcal{T}}^{(\text{sup})} \setminus \{\boldsymbol{u}_q\}, & \text{otherwise.} \end{cases} \tag{23}$$

Then, we update task-specific adapted parameters $\boldsymbol{\theta}_{\mathcal{T}}$ by taking R-steps of Adam optimization on the small support set $\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}$, which is similar to (11)–(16) as

$$\boldsymbol{\theta}_{\mathcal{T}}(R) = \boldsymbol{\theta}^* - \alpha \sum_{r=1}^{R} \frac{\hat{\varpi}(r)}{\sqrt{\hat{v}(r)} + \epsilon} \tag{24}$$

where $\hat{\varpi}(r)$ and $\hat{v}(r)$ represent the values of $\hat{\varpi}$ and $\hat{v}$ at the rth iteration, respectively. $\nabla J(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}_{\mathcal{T}}(r)} \mathcal{L}_{\mathcal{T}}(f_{\boldsymbol{\theta}_{\mathcal{T}}(r)}; \boldsymbol{D}_{\mathcal{T}}^{(\text{sup})})$ and $\mathcal{L}_{\mathcal{T}}(f_{\boldsymbol{\theta}_{\mathcal{T}}(r)}; \boldsymbol{D}_{\mathcal{T}}^{(\text{sup})})$ denote the rth iteration cross-entropy loss measure on the support set $\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}$, $1 \le r \le R$. When $r = 1$ and $\boldsymbol{\theta}_{\mathcal{T}}(1) = \boldsymbol{\theta}^*$, the task-specific optimal parameters $\boldsymbol{\theta}_{\mathcal{T}}^*$ is represented as

$$\boldsymbol{\theta}_{\mathcal{T}}^* = \underset{\boldsymbol{\theta}_{\mathcal{T}}(R)}{\arg\min}\, \mathcal{L}_{\mathcal{T}}\left(f_{\boldsymbol{\theta}_{\mathcal{T}}(R)}; \boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}\right). \tag{25}$$

The fine-grained meta-testing algorithm is shown as Algorithm 2.

C. Computational Complexity

We first study the complexity of the hard sample meta-training phase. Suppose the number of iterations is T, and the number of training tasks is D. In each iteration, the time-consuming parts are detailed as follows: Sampling the dataset $\mathcal{M}_i$ in Step 3 of Algorithm 1 costs $\mathcal{O}(n)$, where n is the size of the whole dataset $\boldsymbol{X}_{\mathcal{S}}$; dividing the dataset $\mathcal{M}_i$ in Step 5 costs $\mathcal{O}(n_m)$, where n_m is the size of the task $\mathcal{M}_i$; forward propagation and back propagation in Step 6 cost $\mathcal{O}(d|\boldsymbol{D}_i^{(\text{sup})}|)$ (d represents the parameter dimensionality of the model); computing the loss on $\boldsymbol{D}_i^{(\text{que})}$ in Step 7 requires a cost of $\mathcal{O}(d|\boldsymbol{D}_i^{(\text{que})}|)$; the cost of update $\mathcal{A}$ in Step 8 is $\mathcal{O}(n_m)$; and the cost of training for K iterations for all samples in $\mathcal{A}$ from Step 10 to 14 is $\mathcal{O}(K|\mathcal{A}|)$. Therefore, the computational complexity per iteration is $\mathcal{O}(n + D(n_m + d|\boldsymbol{D}_i^{(\text{sup})}| + d|\boldsymbol{D}_i^{(\text{que})}|) + K|\mathcal{A}|)$. Then, the overall computational complexity of Algorithm 1 is $\mathcal{O}(T(n + D(n_m + d|\boldsymbol{D}_i^{(\text{sup})}| + d|\boldsymbol{D}_i^{(\text{que})}|) + K|\mathcal{A}|))$. Note that T is not > 300, D is not > 7, n_m is not > 200, $|\boldsymbol{D}_i^{(\text{sup})}|$ is not > 20, $|\boldsymbol{D}_i^{(\text{que})}|$ is not > 20, and $|\mathcal{A}|$ is not > 100,; that is $T \ll n$, $D \ll n$, $n_m \ll n$, $|\boldsymbol{D}_i^{(\text{sup})}| \ll n$, $|\boldsymbol{D}_i^{(\text{que})}| \ll n$, $|\mathcal{A}| \ll n$, and the magnitude of d and n is comparable. Therefore, Algorithm 1 can be solved in polynomial time with respect to the number of samples.

We then analyze the time complexity of the fine-grained meta-testing phase in Algorithm 2. Sampling the dataset $\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}$ in Step 1 costs $\mathcal{O}(|\boldsymbol{X}_{\mathcal{T}}|)$; for the previous c iterations, the cost per iteration in Step 3 is $\mathcal{O}(d|\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}|)$; for the first iteration, calculating Δy_1 requires the cost of $\mathcal{O}(d|\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}|)$, which is the same as the cost required in Step 7 in the cth iteration, and the cost of filtering sample in Step 8 is $\mathcal{O}(|\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}|)$; suppose the number of remaining samples after the cth iteration is n_f, the cost for each of the subsequent $R-c$ iterations is $\mathcal{O}(dn_f)$. Thus, the fine-grained meta-testing phase can incrementally update the parameters with time complexity $O(|\boldsymbol{X}_{\mathcal{T}}| + cd|\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}| + (R-c)n_f)$. In practice, $|\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}|$ is usually small, i.e., $|\boldsymbol{D}_{\mathcal{T}}^{(\text{sup})}| \ll |\boldsymbol{X}_{\mathcal{T}}|$, so the overall complexity can be simplified as $\mathcal{O}(|\boldsymbol{X}_{\mathcal{T}}|)$, which is not a big issue for the practical operation time.

V. Experimental Results and Analysis

In this section, we utilize data from two publicly available datasets to test the performance of the proposed HSML framework, comparing it with state-of-the-art NLOS identification algorithms.

A. Experiment Setup

Dataset 1[1]*:* Measurements were taken on 7 different indoor scenarios, namely, *office 1*, *office 2*, *a small apartment*, *a small workshop*, *a kitchen with a living room*, *a bedroom*, and *a boiler room*. In each scenario, 3000 LOS samples and 3000 NLOS samples were taken. The dataset was created using an SNPN-UWB board with DecaWave DWM1000 UWB radio module. As stated in this public dataset, the CIR data needs to be divided by the number of acquired receiver preamble symbols, a physical parameter output by a device.

Dataset 2[2]*:* This dataset is located in a fully furnished apartment and is used to evaluate NLOS detection algorithms. The data was collected using DecaWave EVK 1000 modules, which are distributed throughout the apartment. The dataset includes CIRs measurements from 57 different transmitter-receiver geometric configurations. For each configuration, at least 100 CIRs were collected. A total of 20 740 CIR samples were collected. The CIR data in this public dataset does not need to be divided by the receiver preamble symbols.

[1]Dataset 1 is available at: https://github.com/ewine-project/UWB-LOS-NLOS-Data-Setewine-project/UWB-LOS-NLOS-Data-Set: Repository with UWB data traces representing LOS and NLOS channel conditions in 7 different indoor locations. (github.com).

[2]Dataset 2 available at: https://github.com/marckolak/uwb-cirs-2018GitHub - marckolak/uwb-cirs-2018: UWB CIRs registered in a fully furnished apartment for NLOS detection algorithms evaluation.

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TABLE I
CNNs FOR META-LEARNING

Layer(type)	Shape
10,111 parameters and 1,635,456 FLOPs	
conv2d(Conv2D)	(15,15,8)
leakyrelu(LeakyReLU)	(15,15,8)
maxpooling2d(MaxPooling2D)	(14,14,8)
conv2d1(Conv2D)	(14,14,4)
leakyrelu1(LeakyReLU)	(14,14,4)
maxpooling2d1(MaxPooling2D)	(13,13,4)
conv2d2(Conv2D)	(7,7,2)
leakyrelu2(LeakyReLU)	(7,7,2)
maxpooling2d2(MaxPooling2D)	(6,6,2)
flatten(Flatten)	(72)
dense(Dense)	(64)
leakyrelu3(LeakyReLU)	(64)
dense1(Dense)	(16)
leakyrelu4(LeakyReLU)	(16)
dense2(Dense)	(2)
softmax(SoftMax)	(2)

We utilize CNN to implement the proposed HSML framework. Our choice of CNN stems from its widespread use in the state-of-the-art MAML-based works. For example, MetaLoc [42] and MTL [22] utilize CNNs. As shown in Table I, the CNN architecture contains three convolutional layers, followed by a LeakyReLU activation function, and 2×2 max-pooling. After flattening the CNN, we use two fully connected layers with LeakyReLU to extract the features. Finally, a fully connected layer outputs the result, which is then normalized using softmax.

Data Processing: Before training, we first process the raw data in the preprocessing stage. The raw CIRs preprocessing steps are outlined as follows.

1) Each piece of data consists of an 1×1016 in-phase and quadrature (IQ)-sampled array. We convert the IQ-sampled array into a real sample array, denoted as R-sampled ($R = \sqrt{I^2 + Q^2}$). The real and imaginary components of the complex IQ sample can be represented in Cartesian coordinates. Next, the Euclidean distance from the origin in the complex plane generates the absolute value of the complex IQ sample, corresponding to the amplitude of the signal.
2) The R-sampled array is further trimmed to 225 samples, with 75 samples preceding the Fp index and 150 samples following it (including the Fp), where Fp represents the index of the first path element detected in the CIRs accumulator.
3) Min-max normalization is applied to each CIR in the dataset. This normalization scales each CIR sample into the range $[0, 1]$ such that the highest value is 1 and the lowest value is 0.
4) Each sample is transformed into a 15×15 format. Then, 100 samples are randomly selected from each scenario, ensuring that the number of LOS samples is approximately equal to the number of NLOS samples.

Since CNN is more suitable for handling 2-D data, we convert the 1-D CIRs data into 2-D data for training. Data normalization, which results in smaller numerical values, helps optimize our model to train and enhance its generalization capabilities. It ensures that the model focuses more on features related to signal-to-noise ratio (SNR) and peak characteristics, rather than absolute signal strength. Absolute signal strength can vary greatly depending on different settings and scenarios (e.g., the average distance between the tag and anchor) and does not necessarily indicate larger errors or LOS/NLOS signal propagation.

B. Experiment Settings and Performance Metrics

All experiments were conducted on a laptop equipped with an Intel i5-12500H 2.5GHz CPU and 16GB RAM, and the algorithms were implemented using PyCharm 2024.2 and Python 3.12.5 programming software.

In the hard sample meta-training phase, the parameters of our framework are set as $\alpha = 0.001$, $\beta = 0.02$, $\gamma = 0.8$, $\mu = 0.25$, $\lambda = 2$, $\tau = 0.5$, $\zeta = 0.1$, $m = 30$, and $K = 10$. In the fine-grained meta-testing phase, the parameters of our framework are set as $\sigma = 0.25$ and $b = 0.7$. We detail reasons for the settings and the influence of these parameters in Section V-C.

We consider the following to interpret the NLOS identification results.

1) *True Positive (TP):* The LOS CIR data are identified as LOS CIR data;
2) *True Negative (TN):* The NLOS CIR data are identified as NLOS CIR data;
3) *False Positive (FP):* The NLOS CIR data are identified as LOS CIR data;
4) *False Negative (FN):* The LOS CIR data are identified as NLOS CIR data.

For binary classification problems, the following four metrics are commonly used to evaluate the identification performance: accuracy, precision, specificity, and F1-Score, which can be calculated as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{26}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{27}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{28}$$

$$\text{F1-score} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}. \tag{29}$$

Accuracy indicates the proportion of correctly predicted samples out of the total samples. Precision indicates the proportion of actual positive samples among those predicted as positive by the model. Specificity defines the proportion of actual negative samples that the model correctly predicts as negative. F1-score is used to comprehensively evaluate the model performance on imbalanced datasets.

C. Experimental Results Analysis

Hyperparameter optimization is a critical aspect of DL models. The hyperparameters mentioned in HSML impact the NLOS identification accuracy and the performance of the model. The step size α and β influence the convergence speed and effectiveness of the model, and the threshold for sample screening γ and μ affect the model's ability to distinguish

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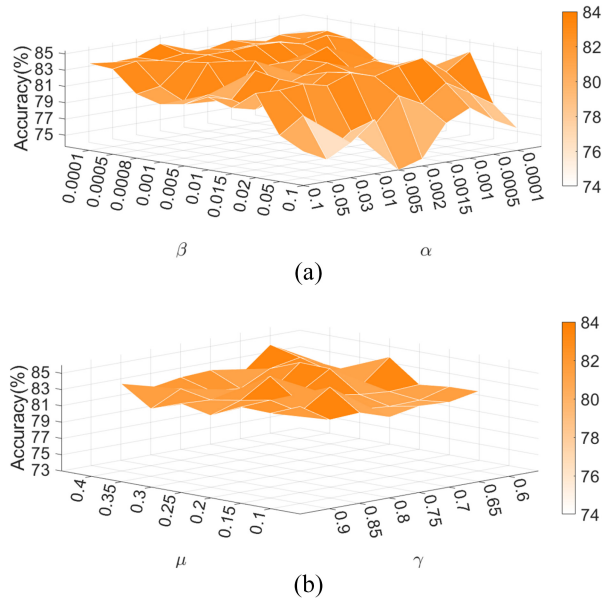


Fig. 4. Effect of parameters on overall identification accuracy. (a) α and β. (b) γ and μ.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT LOSS FUNCTIONS AND PARAMETERS ON DATASET 1

Loss Function	λ	Accuracy (%)
Binary Cross-Entropy (BCE)	N/A	82.2
Mean Squared Error (MSE)	N/A	81.6
Focal Loss	1.5	82.6
Focal Loss	2	84.7
Focal Loss	2.5	83.6
Focal Loss	5	83.0
Focal Loss	10	80.8

between LOS and NLOS components, thereby influencing its generalization capability. Focal loss hyperparameters λ, τ, and ζ collectively influence the model's weight on hard samples during the meta-learning phase, ultimately affecting its overall performance. The hard sample number m and the iteration count on $\mathcal{A}$ collectively influence the model's convergence rate. The threshold for sample filtering hyperparameters σ and b during the fine-grained meta-testing phase affects the degree of fluctuation in the test results, namely the standard deviation (STD) of accuracy. In order to show the effectiveness of the hyperparameters, we carefully design several ablation experiments. The results are shown as Fig. 4 and Table II.

Fig. 4(a) shows the results for step size α and β and 4(b) shows the results for γ and μ. Parameters α and β are selected from {0.0001, 0.0005, 0.001, 0.0015, 0.002, 0.005, 0.01, 0.03, 0.05, 0.1} and {0.0001, 0.0005, 0.0008, 0.001, 0.005, 0.01, 0.015, 0.02, 0.05, 0.1}, respectively. Parameters γ and μ are selected from {0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9} and {0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4}, respectively. From Fig. 4(a), it can be observed that α is generally safe to be selected from {0.0001, 0.0005, 0.001, 0.0015, 0.002, 0.005, 0.01, 0.03}, and the result is not sensitive to β within 0.03. Notably, $\alpha = 0.001$ and $\beta = 0.02$ lead to better performance. Besides, Fig. 4(b) indicates that different threshold combinations have minimal impact on identification accuracy.

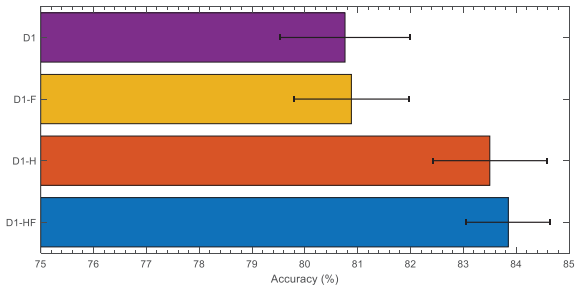


Fig. 5. Test results under different conditions.

We compare different loss functions and different values of hyperparameters on the performance of our framework and show the result in Table II. Binary cross-entropy (BCE) and mean squared error (MSE) are the most commonly used loss functions in classification problems. The formula BCE is shown as (7), and that of MSE can be found in [43].

We set the weight of the hard sample hyperparameter λ of focal loss as {1.5, 2, 2.5, 5, 10}. Our framework performs better with focal loss function rather than BCE or MSE. Choosing an appropriate value for λ can improve the framework's performance. If λ is set too low, it may reduce the framework's effectiveness, while an excessively high value can lead to overfitting, which also lowers classification accuracy.

As described in Section IV, two major modules in the proposed HSML framework contribute to enhancing the identification accuracy, i.e., hard sample meta-training and fine-grained meta-testing. To evaluate the contribution of each model, we conducted ablation experiments on four combinations with and without these two modules. We trained the HSML framework on 600 randomly sampled samples across 6 scenarios. Then, we randomly selected 20 samples from the dataset within the remaining scenario as the support set, and used the remaining samples as the query set. After performing fine-grained meta-testing on the support set, we evaluated the model's performance on the query set.

Fig. 5 compares experimental results under different conditions, namely, D1, D1-F, D1-H and D1-H. D1 represents dataset 1 without data fine-grained meta-testing or hard sample meta-training, D1-F represents dataset 1 including data fine-grained meta-testing but no hard sample meta-training, D1-H represents dataset 1 including hard sample meta-training but no data fine-grained meta-testing and D1-HF represents dataset 1 including data fine-grained meta-testing and hard sample meta-training. We take the average overall accuracy and STD of 100 Monte Carlo experiments as the results. As shown in Fig. 5, D1-HF achieves overall accuracy of $83.85\% \pm 0.40\%$, D1-H reaches $83.50\% \pm 0.54\%$, D1-F reaches $80.82\% \pm 0.44\%$, and D1 reaches $80.76\% \pm 0.62\%$.

It is clear that incorporating the hard sample module contributes to a higher average identification accuracy compared to its absence. Additionally, incorporating the fine-grained meta-testing reduces the STD of identification accuracy. A combination of both enables our HSML model to achieve the highest average identification accuracy along with the smallest

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TABLE III
PERFORMANCE METRICS FOR DIFFERENT COMBINATIONS

Target scenario*	Mean (%)	STD (%)	Min (%)	Max (%)
Office 1	83.16	0.70	81.80	83.80
Office2	83.63	0.83	81.90	84.20
Small apartment	84.83	0.56	83.80	85.30
Small workshop	84.34	0.61	83.40	85.10
Kitchen	84.31	0.78	82.70	85.00
Bedroom	83.76	0.62	82.50	84.60
Boiler room	82.74	0.78	81.40	83.80

∗ For each target scenario, we use the other six scenarios to train the HSML model.

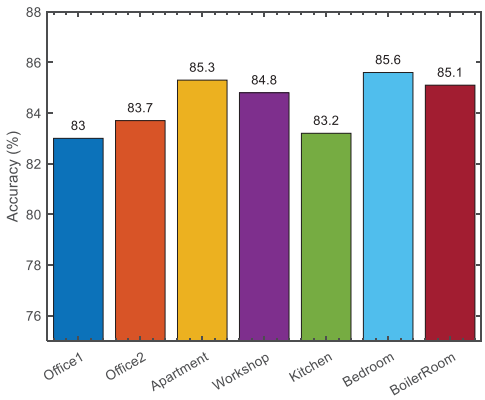


Fig. 6. Test results of each source domain scenario combination.

standard deviation. Notably, the accuracy with both modules is higher than with only the hard sample module, and the standard deviation is smaller than with only fine-grained meta-testing. This indicates a synergistic effect between the two modules, where they complement and enhance each other's contributions to both accuracy and stability.

Based on the above analysis, we conduct further experiments to capture the relationships between data from different scenarios and the HSML performance. We select six out of seven scenarios as the source domains, with the remaining one as the target domain, resulting in seven different combinations. For each combination, we test the model on all seven scenarios and calculate the mean, STD, and extreme values of the combination, as shown in Table III for analysis. Moreover, we individually display the performance of the meta-parameters in the target domain for each combination in Fig. 6. In the HSML framework, we use hard sample meta-training and fine-grained meta-testing, achieving an identification accuracy of 85.6%, which is considered quite reliable in practice. As shown in Table III, the accuracy for each combination varies within a range of 2.5%, whereas the range for the mean accuracy across different combinations is 2.1%. This indicates that the meta-parameters obtained from HSML yield stable test results across different scenarios. Although the data distributions vary across scenarios, leading to some fluctuation in the test results, the variations are minor. This demonstrates that our proposed framework has strong generalization and robustness. Moreover, the range of the target domain test results under different combinations in Fig. 6 is also 2.6%, further validating this point.

D. Comparative Analysis of the Proposed Approach

In order to evaluate the NLOS identification performance of the proposed HSML framework, we consider the following comparative algorithms.

Fuzzy Credibility-Based SVM (FC-SVM) [17]: FC-SVM identifies NLOS components based on combinations of statistical features extracted from CIR. Such features include kurtosis, root mean square delay spread, power difference, energy of the received signal, rise time, etc.

FCN-Attention [27]: Building on the CNN, a self-attention mechanism is incorporated to analyze the relationships between CIR components, thereby improving the accuracy of NLOS identification.

CNN and Long Short-Term Memory (CNN-LSTM) [44]: LSTM is integrated into the CNN for data preprocessing. Additionally, embedded AI technology and a quantized TinyML approach are implemented to strengthen the robustness of NLOS recognition.

TL [16]: TL generates samples by using GAN and few collected target domain CIR data. Then, TL is utilized to reduce the distribution divergence between the source and target domains.

FineTune [45]: FineTune is an algorithm based on CNN and TL, which achieves cross-scenario NLOS identification by training CNN parameters on the source domain and fine-tuning them on the target domain.

We compare our HSML framework with the above-mentioned methods on Dataset 1 and Dataset 2. When comparing on dataset 1, we merge the data from the six source domains into one dataset for training and test the results in the target domain. When comparing on dataset 2, since the amount of NLOS and LOS data are completely unequal, indicators such as identification accuracy cannot accurately characterize the performance of the model. Therefore, we randomly sample the NLOS data to make the number of NLOS samples roughly the same as that of LOS samples, and then test the model. Fig. 7 shows the results of the test on dataset 1. The performance of each algorithm under different scenarios in terms of the metrics accuracy, precision, specificity, and F1 score is shown in Fig. 7(a)–(d), respectively.

As shown in Fig. 7, the proposed HSML framework exhibits an average accuracy of 84.0%, which is 10.5%, 4.3%, 6.1%, 4.7%, and 5.4% higher than FC-SVM, CNN-LSTM, FineTune, FCN-Attention, and TL, respectively. FC-SVM, FineTune, and TL achieve lower accuracy in cross-scenario NLOS identification, while CNN-LSTM and FCN-Attention exhibit better performance in terms of F1 Score in each scenario, indicating that they maintain a good balance between positive and negative samples in imbalanced datasets. The FC-SVM algorithm does not consider cross-domain information or the impact of CIR data collection using different devices and the calculation of physical parameters, resulting in lower identification accuracy in new environments. Furthermore, because the source domain data comes from multiple environments, FineTune struggles to effectively transfer knowledge from the source domain to the target domain, resulting in poor performance of the model on the target domain. On the other hand, TL depends

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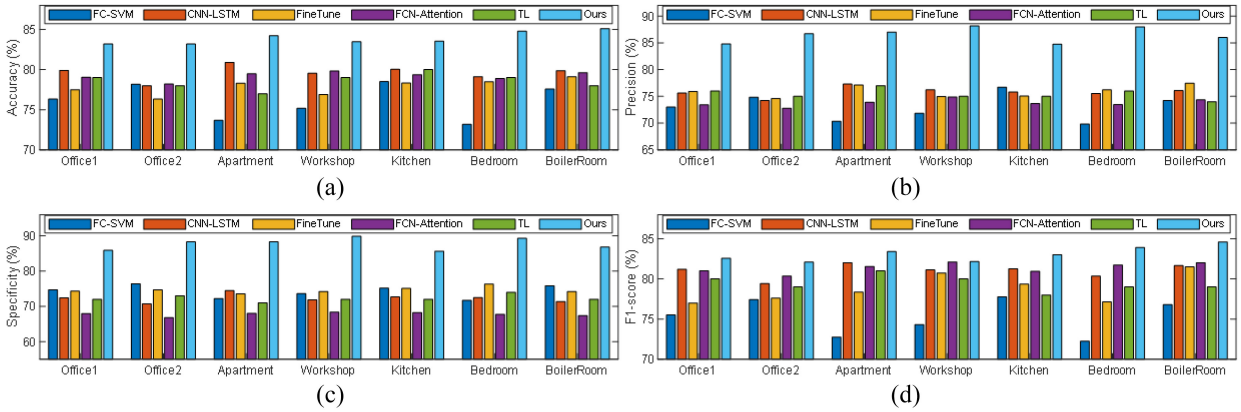


Fig. 7. Results of different algorithms in dataset 1. (a) Accuracy. (b) Precision. (c) Specificity. (d) F1 score.

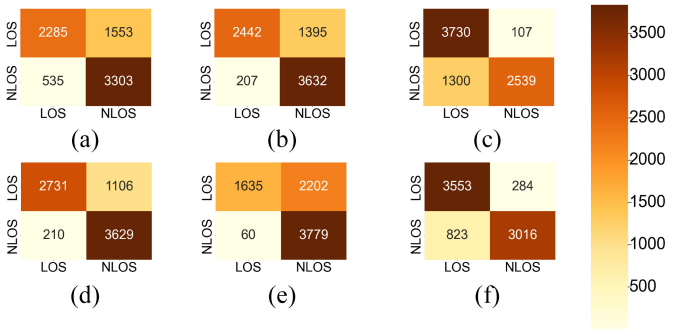


Fig. 8. Confusion matrices of various algorithms on dataset 2. (a) FC-SVM. (b) FineTune. (c) CNN-LSTM. (d) FCN-Attention. (e) TL. (f) Ours.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON DATASET 2

Model	Accuracy (%)	Our model improved by (%)
FC-SVM	72.80	17.55
FineTune	79.13	8.15
CNN-LSTM	81.67	4.79
FCN-Attention	82.86	3.28
TL	70.53	21.34
Ours	**85.58**	N/A

on target domain data and GANs to generate domain-specific data. However, when the sampled data from the target domain is not sufficiently representative, or if the GAN structure is inadequately designed, the computed features may fail to fully capture the target domain's characteristics, leading to reduced identification accuracy. FCN-Attention effectively extracts the relationships among the internal components of each sample, enabling it to capture the intrinsic features of LOS and NLOS samples and better performance across accuracy and F1 score. However, FCN-Attention algorithm does not consider the feature differences of LOS and NLOS samples across different scenarios, severely limiting its accuracy. In contrast, our proposed HSML algorithm is capable of capturing both cross-scenario generalization properties and scenario-specific features, resulting in enhanced performance and robustness when tested in a new scenario.

Furthermore, we present the test results of the comparative models on Dataset 2. As mentioned in Section V-A, Dataset 2 was collected in a new scenario using DecaWave EVK 1000, which is different from Dataset 1 and significantly increases the difficulty of NLOS identification. To visually demonstrate the performance of each model, we present the confusion matrices of comparative methods in Fig. 8. Confusion matrices are effective tools for evaluating the performance of identification algorithms. In Fig. 8, rows represent the actual labels, and columns represent the predicted labels.

As shown in Fig. 8, the proposed method demonstrates clear superiority over the other methods—FC-SVM, FineTune, CNN-LSTM, FCN-Attention, and TL, in a new scenario. HSML's confusion matrix reflects a better balance, with fewer false LOS identification and false NLOS identification. Other models, while performing adequately, tend to misclassify more NLOS cases as LOS or vice versa. The identification accuracy of the TL method is especially lower than that of FC-SVM. This is because the TL method requires a large amount of training data to align feature distribution across different domains, which cannot be realized in the few sample scenarios. Furthermore, to intuitively show the performance gap, we present the accuracy of the comparative methods in Table IV. Notably, the proposed HSML outperforms all the state-of-the-art methods in terms of accuracy. Hence, we conclude that the proposed HSML method exhibits improved performance and robustness across different scenarios and domains.

E. Complexity Analysis

As shown in Table I, there are 10 111 parameters in the model and 1 635 456 FLOPs for every sample in one iteration, while the CPU can calculate 672 GFLOPs, which is much higher than the computing resources required for NN iterations. The average time consumed per iteration during the hard sample meta-training phase is 55 s, thus limiting the computation overhead to a negligible amount. When testing in a new scenario, only 20 samples are needed in the fine-grained meta-testing phase, which is significantly lower than the number of samples required during the hard sample meta-training phase, and the number of iterations is expected to be lower. Therefore, the application of our proposed method is not affected by computation resource constraints.

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VI. Conclusion

In this article, we have proposed an HSML framework for cross-scenario and cross-domain UWB NLOS identification. The proposed framework mainly includes a hard sample meta-training phase and a fine-grained meta-testing phase. In the first phase, the cross-scenario and cross-domain NLOS identification problem is modeled as a series of learning tasks. Then, we train a two-loop NN, where the inner loop learns the specific feature of each task and the outer loop learns the cross-task generalization properties. Moreover, the proposed hard sample meta-training focuses on leveraging hard samples, thereby enhancing the model's robustness and generalization properties. In the fine-grained meta-testing phase, samples are filtered based on the tendency of estimated residuals during fine-tuning, reducing the risk of poor performance caused by anomalous samples. Experiment results on different datasets show that the proposed method outperforms other models in terms of identification accuracy, robustness, and generalization performance. Future work will consider building a foundation NLOS identification model to assist high-accuracy UWB positioning in unseen scenarios.

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Yinong Liu received the B.Eng. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 2022, where he is currently pursuing the M.Eng. degree in communication.

His current research interests include indoor localization, machine learning, federated learning, and transfer learning.



Haonan Si (Graduate Student Member, IEEE) received the B.Eng. degree from North University of China, Taiyuan, China, in 2021. He is currently pursuing the Ph.D. degree with the Department of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu, China. His research interests include indoor localization, machine learning, multiagent systems, and nonlinear tracking control.



Gordon Owusu Boateng (Member, IEEE) received the bachelor's degree in telecommunications engineering from the Kwame Nkrumah University of Science and Technology, Kumasi, Ghana, in 2014, and the M.Eng. and Ph.D. degrees in computer science and technology from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2019 and 2023, respectively.

From 2014 to 2016, he was with SubContracts at Ericsson, Accra, Ghana, and TIGO, Accra. He is currently a Postdoctoral Researcher with the Hybrid Positioning Research Group, UESTC. Till now, he has published over 40 scientific journals and conference papers. His research interests include 5G/6G wireless networks, blockchain, reinforcement learning, vehicular networks, target positioning, and automated valet parking.



Xiansheng Guo (Senior Member, IEEE) received the B.Eng. degree from Anhui Normal University, Wuhu, China, in 2002, the M.Eng. degree from Southwest University of Science and Technology, Mianyang, China, in 2005, and the Ph.D. degree from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2008.

From 2008 to 2009, he was a Research Associate with the Department of Electrical and Electronic Engineering, University of Hong Kong, Hong Kong. From 2012 to 2014, he was a Research Fellow with the Department of Electronic Engineering, Tsinghua University, Beijing, China. From 2016 to 2017, he was a Research Scholar with the Advanced Networking Laboratory, Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, NJ, USA. He is currently a Full Professor with the Department of Electronic Engineering, UESTC. His research interests include array signal processing, wireless localization, machine learning, information fusion, and unmanned driving.



Yu Cao (Graduate Student Member, IEEE) received the B.Eng. degree from Ningbo University, Ningbo, China, in 2023. He is currently pursuing the M.Eng. degree in communication with the University of Electronic Science and Technology of China, Chengdu, China.

His current research interests include indoor localization, machine learning, federated learning, and transfer learning.



Bocheng Qian received the B.Eng. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 2021, where he is currently pursuing the Ph.D. degree with the Department of Information and Communication Engineering.

His current research interests contain machine learning, multimodal data fusion, indoor localization, and magnetic field positioning.



Nirwan Ansari (Life Fellow, IEEE) received the B.S.E.E. degree (summa cum laude with a perfect GPA) from New Jersey Institute of Technology (NJIT), Newark, NJ, USA, the M.S.E.E. degree from the University of Michigan at Ann Arbor, Ann Arbor, MI, USA, and the Ph.D. degree from Purdue University, West Lafayette, IN, USA.

He is a Distinguished Professor of Electrical and Computer Engineering, NJIT, He authored *Green Mobile Networks: A Networking Perspective* (Wiley-IEEE, 2017) with T. Han, and co-authored two other books. He has also (co-)authored over 700 technical publications, with more than half of them published in widely cited journals and magazines. His current research focuses on green communications and networking, edge computing, drone-assisted networking, and various aspects of broadband networks.

Dr. Ansari's many recognitions are several excellence in teaching awards, multiple best paper awards, the NCE Excellence in Research Award, several ComSoc TC technical recognition awards, the NJ Inventors Hall of Fame Inventor of the Year Award, the Thomas Alva Edison Patent Award, the Purdue University Outstanding Electrical and Computer Engineering Award, the NCE 100 Medal, the NJIT Excellence in Research Prize and Medal, and designation as a COMSOC Distinguished Lecturer. He was elected to serve for the IEEE Communications Society Board of Governors as a Member-at-Large. He has served as the Director for ComSoc Educational Services Board, chaired various technical and steering committees within ComSoc, and has served on many committees, such as the IEEE Fellow Committee. He has actively participated in organizing numerous IEEE International Conferences/Symposia/Workshops. He has also been granted more than 40 U.S. patents. He has served as a Guest Editor for numerous special issues on various emerging topics in communications and networking. He currently serves as an Editor-in-Chief for IEEE WIRELESS COMMUNICATIONS and has been on the editorial/advisory board for over ten journals. He is a Fellow of the National Academy of Inventors.

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